Tencent HY

Space Track: How Multimodal Agents Perceive, Update, and Act in Changing 3D Worlds

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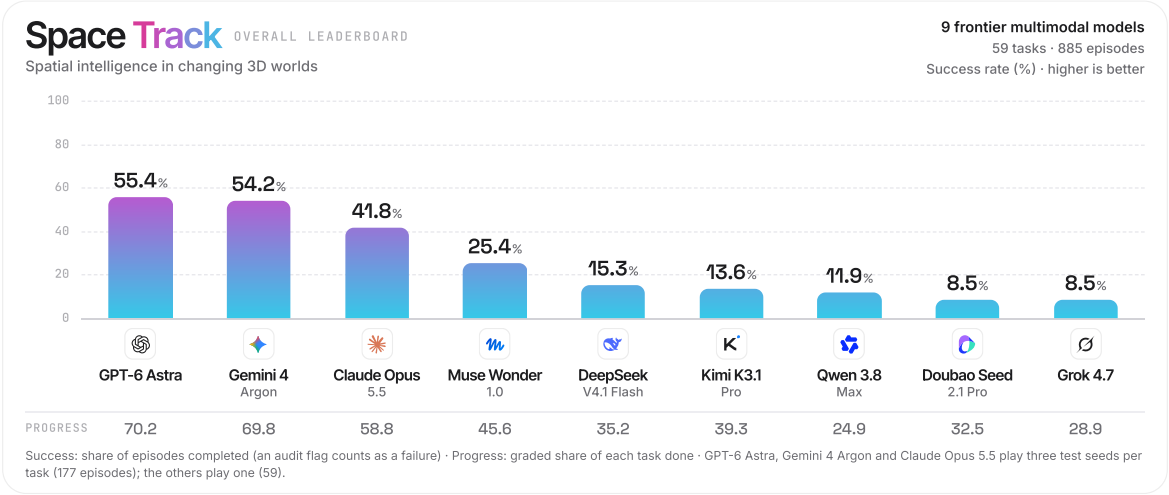

Figure 1: **Overall leaderboard of Space Track.** Nine frontier multimodal models on all 59 tasks, ranked by success rate; progress is the graded share of each task done.

Abstract

Humans do not merely remember the spaces they have seen; they keep track of spaces as they change—after knocking down a wall, flipping a breaker, or noticing someone else walking the same corridors—and they act on that updated picture. Can multimodal agents do the same? We present Space Track, a closed-loop benchmark of five walkable first-person 3D worlds that change while the agent acts: through the agent's own construction work, through switches that rewire a floor with non-Euclidean topology, through a pursuer patrolling a seven-level sanatorium by published rules, through reference frames that shift between the writer and the reader of a note, and through obstacles moving on fixed cycles. Its 59 tasks span tens to hundreds of steps and include 15 counterfactual controls that each isolate a single source of difficulty. Every test seed is verified by an oracle, every episode runs on an isolated, audited evaluation server, and every task is playable by humans through the same browser interface. Evaluating nine frontier models over 885 episodes and 46k actions reveals a steep capability cliff: GPT-6 Astra and Gemini 4 Argon complete 55% and 54% of tasks, the remaining seven reach at most 42%, and in the detective's office—where reference frames shift—six of the nine fail every task. Trajectory and self-explanation analyses show that perceiving change is not the bottleneck. On a floor whose stairs loop back on themselves, failing agents accurately describe the loop in their own words, yet 15 of 18 failures on the two topology-dependent tasks end with a reachable goal declared impossible; the same tasks succeed in 80% of episodes on an ordinary floor versus 40% on the non-Euclidean one. Across worlds, agents misjudge termination, declaring unfinished renovation work complete in 48% of episodes and abandoning feasible corridor, sanatorium, and office tasks as impossible in 21%. Providing a writable workspace improves six of nine models, and supplying the ground-truth map nearly doubles success on the main tasks (from 25% to 47%, eliminating 75% of corridor failures), while adding a correct verbal plan raises success to 55%—demonstrating that spatial intelligence in changing worlds hinges on closed-loop model revision and calibrated execution rather than static perception alone.

1 Introduction

When we renovate a flat, we know the moment a wall comes down that the kitchen is now one step from the sofa. Lost in a basement whose corridors seem to repeat, we chalk a mark on the wall and watch for it to come round again. Walking a dark building where someone else is on patrol, we keep a rough sense of where they are and which way leads out. We live in a world that changes as we act in it, and much of our spatial intelligence goes into keeping our picture of that world current: perceiving what is there, updating what we know when it changes, and acting on the updated picture (Tolman, 1948; Baddeley, 1992; Newcombe, 2024).

Multimodal large language models (MLLMs) are increasingly asked to act as embodied agents, and a growing body of work asks whether they can "think in space". Recent benchmarks show that MLLMs can see, remember and recall the layout of a room from video with emerging but subhuman competence (Yang et al., 2025a), struggle to imagine unseen views (Wang et al., 2026a), lose accuracy as a stream of observations grows (Lin et al., 2025), and build less accurate beliefs when they must gather the evidence themselves (Zhang et al., 2026a). In all of these settings, however, the space holds still. Interactive benchmarks close the loop between perception and action, but in homes whose structure is fixed and whose progress is observable (Shridhar et al., 2020; Yang et al., 2025b), or from top-down views of objects being edited (Liu et al., 2026). The regime embodied agents actually face, in which the world changes while they act, is seen only from a first-person view, and nothing on screen marks what has changed or what is finished, remains largely unmeasured.

To study spatial intelligence where the world changes, we present Space Track (Figure 2). It consists of five walkable first-person worlds, each built around a different source of change. In a **renovation** flat, the agent changes the layout itself: it cleans, paints, installs fixtures and knocks down a wall, with no progress bar to tell it what is done. In an **impossible corridor**, a ring that returns to the lift lobby after three right turns and stairs that lead back to the floor they left, breakers, fire shutters and one-way doors rewire which corridors connect. In a hillside **sanatorium** of four buildings and seven levels, a night matron patrols, listens and searches by published rules. In a **detective's office**, a note written from the detective's chair must be read from the doorway, and opening the wrong sealed box ends the episode. On an **obstacle course**, hammers, rafts and ropes move on cycles that the agent must infer from bursts of frames. All five worlds share one interface, and every task can be played by a person in the browser through the same page the models act on.

Three design principles make Space Track diagnostic rather than merely hard. **No scaffolds:** agents receive first-person frames and a one-line report of what their last action did, and nothing else, with no map, progress bar, highlight or object identifier; every tool acts only on what lies under the crosshair within reach. **Counterfactual controls:** 15 of the 59 tasks are controls that share seeds with a main task and remove exactly one difficulty, such as an ordinary floor in place of the impossible one, so that the task–control gap isolates the skill under test. **Trustworthy evaluation:** an isolated server holds the environment, the private seed bank and the scores; on every test seed an oracle completes the task within the step budget, or correctly declares it impossible where it cannot be done; and trajectories are audited for shortcuts such as declaring a task impossible without having seen the obstacle.

Evaluating nine frontier models through a unified agentic harness (§4; Figure 1), we find a steep capability cliff: GPT-6 Astra and Gemini 4 Argon complete 55% and 54% of the tasks, Claude Opus 5.5 reaches 42%, and the remaining six complete 25% or fewer; different worlds favour different models, and shifting reference frames in the detective's office poses the steepest barrier. Analysing trajectories and the reasons agents state for their terminal decisions (§5), we find that agents reliably perceive local changes yet fail to act on their global implications. On the non-Euclidean floor, they record the looping stairwell in their own words and then declare a reachable goal impossible; across worlds, they stop with work unfinished and abandon feasible tasks while obeying explicit constraints almost without fail. Finally (§6), we test three forms of intervention: a writable workspace improves six of nine models, counterfactual controls isolate the

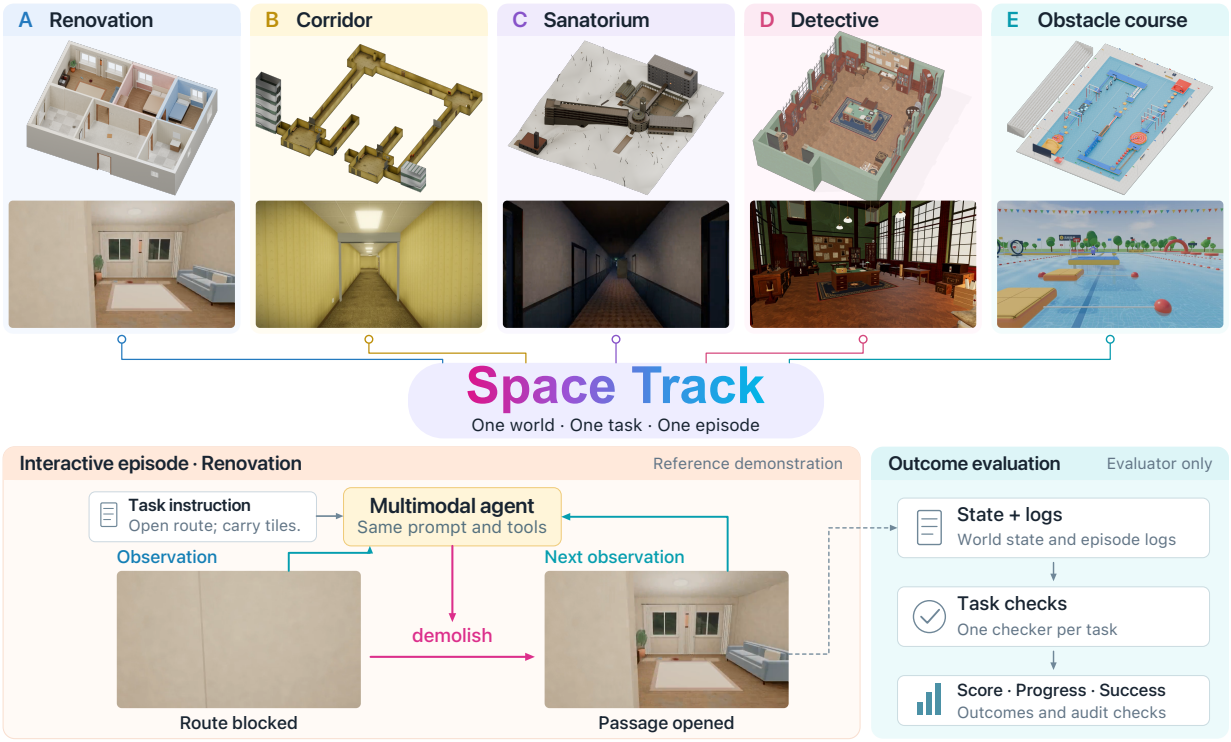

Figure 2: **Five worlds, one interface.** Top: the five worlds of Space Track, each with a first-person view. Bottom left: one interactive episode in the renovation. The agent receives the task instruction and a first-person observation, acts (here `demolish`), and the next observation shows the change it made. Bottom right: the world state and episode logs are visible only to the evaluator, whose task-specific checks produce the score, progress and success, and apply the audit checks.

cost of state revision and completion tracking, and stratified assistance shows that while a ground-truth map nearly doubles success, 59% of baseline failures persist even when a correct verbal plan is provided.

Contributions.

- A taxonomy of spatial intelligence for acting in changing worlds, structured around the perceive–update–act loop and five distinct channels of environmental change (§2).
- Space Track, a closed-loop benchmark of five first-person 3D worlds and 59 tasks (including 15 counterfactual controls), backed by oracle-verified private seeds, an audited evaluation server, and a shared browser interface for human and model play (§3).
- A comprehensive evaluation of nine frontier models paired with trajectory-level failure analysis and stratified interventions, establishing that the primary bottleneck lies not in perceiving spatial change, but in revising the world model, executing against dynamic elements, and calibrating termination (§4–6).

2 Spatial Intelligence in a Changing World

We scope the capabilities Space Track measures and provide a framework for the analysis that follows.

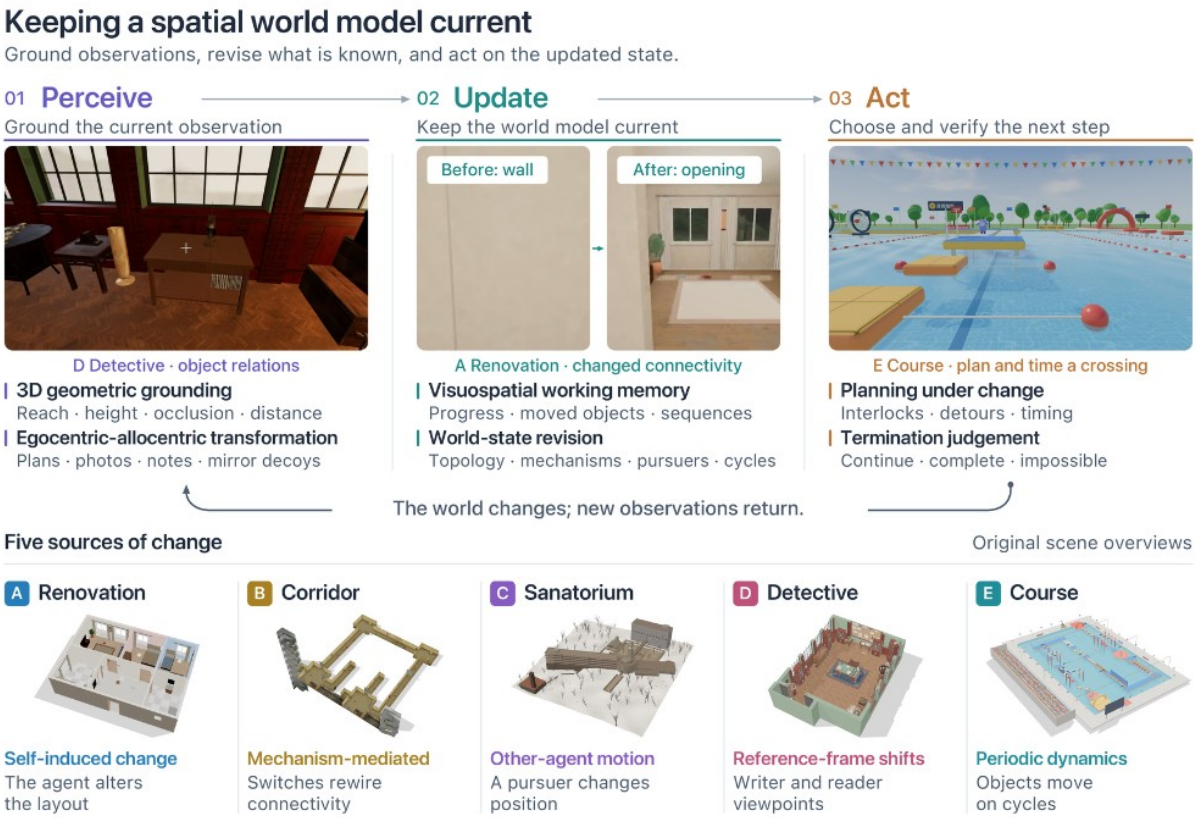

Figure 3: **A taxonomy of spatial intelligence for acting in changing worlds.** Top: six capabilities grouped by the step of the perceive–update–act loop they serve. Bottom: the five sources of change around which the worlds of Space Track are built. Table 1 maps worlds to capabilities.

Scope. Classic tests of spatial ability include pen-and-paper tasks such as mental rotation (Shepard and Metzler, 1988). Our focus is spatial intelligence as it is exercised while acting in building- and room-scale 3D spaces: walking, looking, operating things and deciding when a task is done, from a first-person view. We refer to the agent's internal picture of the space and its state as its *world model*, in the sense of a mental model (Johnson-Laird, 1983) that supports both inference and action.

Taxonomy. Figure 3 organises the capabilities involved along the loop an agent runs at every step. To **perceive**, the agent must ground what it sees in 3D geometry, including reach, height, occlusion and distance, in the sense of the affordances a scene offers to a body (Gibson, 1979), and it must transform between egocentric views and allocentric descriptions such as a floor plan, a photo or a note written from someone else's position. To **update**, it must hold the state of the world in visuospatial working memory (Baddeley, 1992), such as which walls are painted and which furniture it moved, and revise its model when the world changes: a wall comes down, a shutter drops, a pursuer moves, a stairwell turns out to repeat. To **act**, it must plan under the current model, routing around interlocks, timing a jump or a dash, and judge when to terminate, stopping only when the task is done and declaring it impossible only when no route exists.

Sources of change. Change enters a world through different channels, and each places a different demand on the update step. It can be *self-induced*, when the agent's own actions alter the layout or the appearance of landmarks (A); *mechanism-mediated*, when switches and doors the agent operates rewire connectivity (B);

Table 1: **Capabilities stressed by each world** (● primary, ○ secondary), by design.

	Perceive		Update		Act	
World	3D grounding	Ego–allo	Working memory	Revision	Planning	Termination
A Renovation	●	●	●	●	○	●
B Corridor	○	○	●	●	●	●
C Sanatorium	○	●	●	●	●	●
D Detective	●	●	●	○	○	●
E Course	●	–	○	●	●	–

Table 2: **Benchmark statistics**, read from the task cards the agents receive. Step budgets are per task (per section on the course).

World	Main tasks	Controls	Action level	Action types	Frames / obs.	Step budget
A Renovation	5	5	parametric	18	1	40–286
B Corridor	8	7	junction	12	1	50–56
C Sanatorium	11	1	junction	11	1	28–116
D Detective	8	2	parametric	16	1	28–109
E Course	12	0	parametric	13	4	19–79
Total	44	15				

exogenous, when another agent moves through the space (C); *referential*, when instructions are given from another viewpoint, light fails or the agent must undo its own traces (D); or *dynamic*, when objects move on their own schedule (E). Table 1 shows how the worlds cover the six capabilities. Every task requires all three steps of the loop, but the emphasis differs: the course stresses geometric grounding and timing, the office reference frames and memory, the corridor and the sanatorium world-state revision and termination.

3 Space Track

3.1 Overview

Space Track evaluates agents in closed loop: at each step the agent receives an observation, issues an action, and the world advances. All five worlds are browser scenes rendered with three.js and driven through headless Chrome with software rendering, so evaluation needs no GPU. Observations are 768×432 first-person frames (a burst of four frames $0.25\,\mathrm{s}$ apart on the course, where motion must be inferred) and a one-sentence report of what the action did to the agent's body. The world pauses while the agent thinks, and each action advances the game clock by its physical duration, so inference latency does not confound spatial competence (Zhang et al., 2025). Worlds with fine manipulation use parametric actions (`walk(m)`, `turn(deg)`, `look(pitch)`, `jump(m)`, ...) and tools that act under the crosshair; the two building-scale worlds use junction-level navigation (`forward` to the next junction, door or stair; `turn` by $45°$) with `use`, `interact` and `wait`. In the flat, the corridor, the sanatorium and the office, the agent ends an episode with `stop` or `declare_impossible(reason)`; the flat, the office and the course also accept `give_up`, and a course section otherwise ends when it is passed. Table 2 summarises the worlds, and Figure 4 illustrates representative tasks and their spatial demands across all five worlds.

A. Renovation. A six-room flat and a client's work order: clean three rooms with no progress indicator, paint "the bedroom next to the bathroom" including the wall behind the wardrobe, install a radiator under

every window, knock down the kitchen–living wall (not the load-bearing one, marked only on the attached floor plan) and carry tiles through the new opening, or repaint a room and then find it again.

B. Corridor. A basement floor whose ring corridor returns to the lift lobby after three right turns, and whose west stairs lead down to the same floor again. A breaker panel powers at most two circuits, two fire shutters are interlocked, a pump drains a flood over time, and some doors open one way. Tasks provide a 50 m safety rope, chalk, or a camcorder with six battery bars of night vision and playback; two add a pursuer slightly faster than the agent.

C. Sanatorium. A hillside tuberculosis sanatorium of a 1934 wing, a 1992 wing whose second floor is level with the old wing's third, a rotunda, service buildings and a mortuary tunnel to the boiler house, with 33 independent loops and 19 hiding places. The night matron hears footsteps along walking distance, sees a lit flashlight at 30 m and searches lockers she saw the agent enter; her rules are given to the agent.

D. Detective. One densely furnished office with 28 interactive props and 11 sealed boxes and bookcases. The detective's note is written from her chair, the next clue from the doorway, each mirrored by a decoy. Other tasks: find three objects moved after dark with a hand lantern, copy documents and push the furniture back, return relics, and judge whether a high display case is locked.

E. Course. Twelve sections over a pool: hammers with periods of 2.0–3.6 s, a rotating sweeper, large balls whose last gap may exceed the maximum jump, a punch wall firing in a pattern, a reverse conveyor, moving rafts, cracked false tiles, crosswind, rope swings and a memory bridge. Motion is periodic and fully determined by the seed.

3.2 Benchmark Construction

Single source of truth. Each world is defined once, by a layout file holding geometry, navigation graph, mechanisms and timings. The browser scene, the optional Blender Cycles renderer (used for photoreal renovation frames with identical rules) and the design drawings are all compiled from it, so what the agent sees, what the oracle plans on and what the scorer checks cannot drift apart.

Oracle-verified seeds. Seeds randomise what an agent could otherwise memorise: stains, colours and the target colour in the flat; which box the note means, what was moved and whether a case is locked in the office; periods, phases, punch order, false tiles and fans on the course. Every world ships an oracle that plans in the same physics as the agent: on a 10 cm grid with exact aiming in the flat, on the navigation graph in the buildings, and by searching over waiting times on the course. A seed enters the dev or private test bank only if the oracle completes every task on it within the step budget, which for a task that cannot be done means declaring it impossible with the correct reason; in the sanatorium the oracle must also survive with the matron walking. A task and its control share seed indices, so the same index is the same world in both. Step budgets are set from the oracle ($3\times$ its step count $+10$ in the flat and on the course).

Playtesting closes shortcuts. Because people play the same pages, playtesting exposes shortcuts that a benchmark must not reward, and each is closed in the layout. On the course, for example, a jump that should be impossible turned out to be reachable from a lower platform, so the landing was raised; walking the seam between two columns of false tiles avoided them, so the columns were separated by half a metre; false tiles that took time to collapse could be crossed quickly, so they now give way on contact.

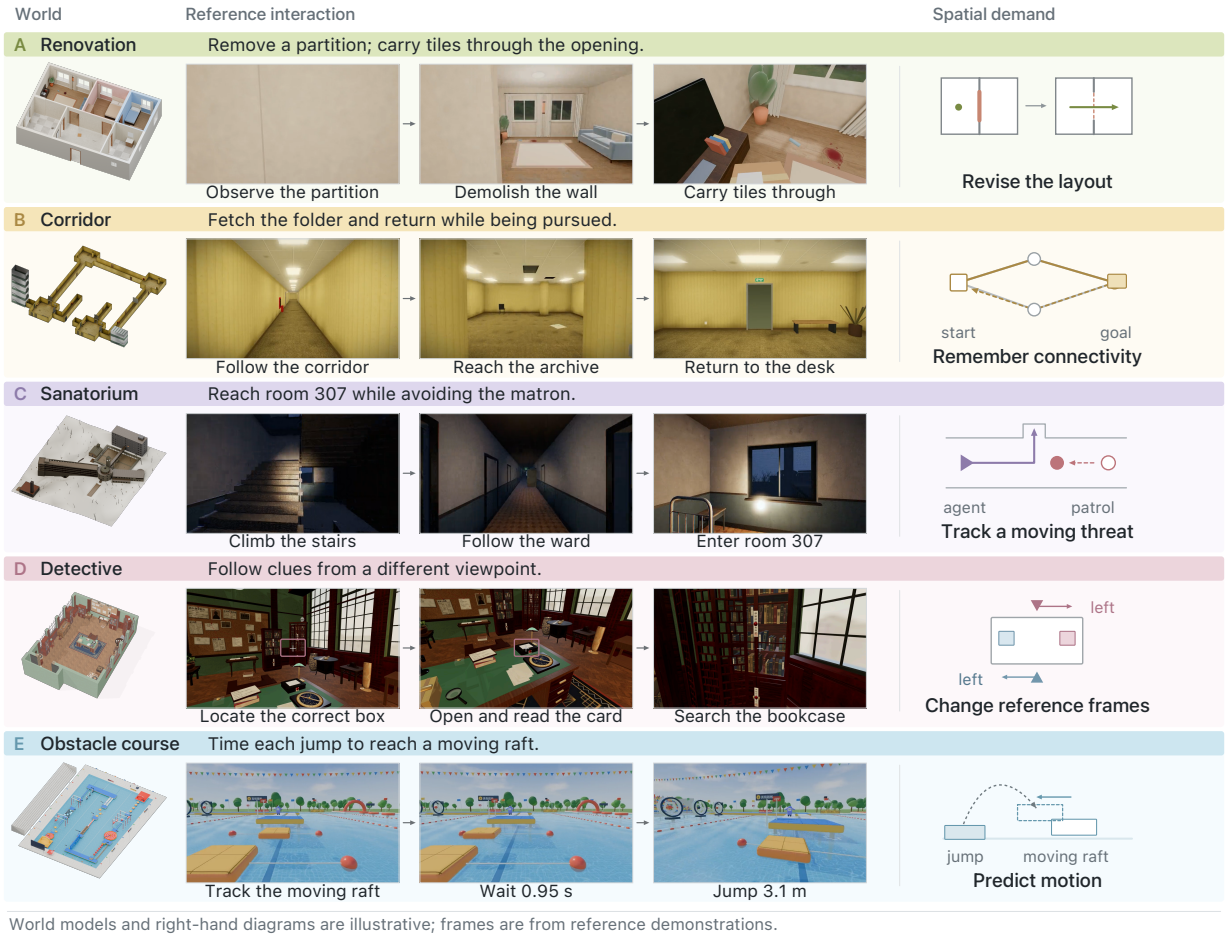

Figure 4: **Tasks and interaction in five worlds.** Whether renovating a flat, finding a way through a floor that does not add up, or slipping past a patrol in a dark building, acting in space means perceiving the scene, updating one's picture of it as it changes, and acting on that picture. Each row shows one world of Space Track (left), three first-person frames from a reference demonstration of one of its tasks (centre), and the spatial demand the task places on the agent (right): revising a layout the agent changed itself, remembering connectivity, tracking a moving threat, changing reference frames, and predicting motion.

Audited evaluation. The evaluation server is the only process that touches the environment, seeds, positions, oracle data and scores; agents see only `create`, `step` and `end`. Every step is logged with the full state, and four audit checks set an episode's score to zero: declaring impossible without having seen a blocker, naming a blocker that was not seen, claiming a clue that was never displayed, and six or more invalid actions.

3.3 Counterfactual Controls

Fifteen controls each remove one difficulty from a main task and keep everything else, including the seed. In the flat, they add a per-room progress hint, replace a relational reference with a door plate, mark installation points on the plan, start with the wall already down, or identify the repainted room by its plate. Seven corridor tasks run on an ordinary floor plan. In the sanatorium, an unreachable fourth floor is paired with a reachable fifth floor up the same stairs, so declaring impossible is correct in one and wrong in the other. In the office, the note is written from the agent's side, or the lights are left on. The full list is in Figure 8.

3.4 Metrics

Success and progress. An episode *succeeds* when the task's own checker marks it complete; an episode with an audit flag counts as a failure. Models are ranked by success rate, with ties broken by progress. Because success is a sparse signal over hundreds of steps, we also report *progress*, the graded share of the task done, analogous to how mean relative accuracy grades numerical answers by their proximity to the truth (Yang et al., 2025a). Worlds A and D award partial credit by work-order item or document; on the course, where each task is a single section, progress equals success. In the corridor and the sanatorium, an unfinished episode earns the covered share of the shortest-path distance on the floor's graph to the next goal, plus goals already reached, scaled by 0.9 so that any success outranks any failure. Declaring impossible a task that can be done halves this progress, and an audit flag zeroes it. A model's progress is its mean over tasks (Appendix C).

Behavioural metrics. We also record *false completion* (`stop` before the task is complete), *false infeasibility* (`declare_impossible` on a task that can be done), hazards (load-bearing walls demolished, seals broken, being caught), and efficiency (share of the step budget used).

3.5 Agent Harnesses

Agentic harness (main). Every model plays through the same pinned version of the Codex command-line agent (v0.164.0) with the same configuration and task prompt, each episode in its own isolated sandbox. The game is exposed as two tools, `get_task` and `act`; the agent also has a shell and an empty working directory for notes. Each model runs at its configured reasoning setting (Appendix E) with a 128k output cap, and the 40 most recent frames are kept in context.

Direct harness. Each turn, the model returns one JSON object with a rolling free-text memory of at most 400 characters, a short justification and one action; it sees the task card, a text record of the episode with the last five justifications, and only the current frames.

4 Evaluation on Space Track

4.1 Evaluation Setup

We evaluate nine frontier multimodal models at the high-reasoning configurations listed in Appendix E: Gemini 4 Argon, GPT-6 Astra, Claude Opus 5.5, Muse Wonder 1.0, DeepSeek V4.1 Flash, Kimi K3.1 Pro, Grok 4.7, Doubao Seed 2.1 Pro and Qwen 3.8 Max. All nine models are evaluated across all 59 tasks through the agentic harness on verified private test seeds: the three frontier models exceeding 40% accuracy—GPT-6 Astra, Gemini 4 Argon and Claude Opus 5.5—are evaluated across three independent test seeds per task (177 episodes each) to quantify seed-to-seed stability, and the remaining six are evaluated on the primary test seed (59 episodes each). In total, the evaluation comprises 885 episodes and 46,332 actions. As a chance baseline, a uniformly random policy evaluated across the 27 junction-navigation tasks (corridor and sanatorium) completes 1 of 81 episodes (raw score 0.01).

4.2 Main Results

Figure 5 reports overall and per-world performance. Our key observations are as follows.

A steep capability cliff. GPT-6 Astra (55.4%, 98 of 177 episodes) and Gemini 4 Argon (54.2%, 96 of 177) form a clear top tier; Claude Opus 5.5 follows at 41.8%, and the remaining six models complete between

Figure 5: **Evaluation on Space Track** (agentic harness, 885 episodes). Left: success rate per world. Right: overall success rate and progress (the graded share of each task done) over all 59 tasks. Models are ranked by success rate. GPT-6 Astra, Gemini 4 Argon and Claude Opus 5.5 play three test seeds per task, the other models one. Table 7 in the appendix adds step-budget use.

8.5% and 25.4% of the tasks. Success sits well below progress for every model: GPT-6 Astra covers 70% of the graded work but completes 55% of tasks, Muse Wonder 1.0 46% and 25%.

Worlds reward different models. GPT-6 Astra leads the corridor (96%), the sanatorium (28%) and the course (86%); Gemini 4 Argon leads the renovation (43%) and the office (33%). The corridor and the course are the most polarised worlds: the top three complete 62–96% of corridor tasks and 64–86% of course sections, the other six at most 27% and 42%.

Switching reference frames is hardest. The detective's office, where notes are written from another viewpoint and the room must be remembered in the dark, is the hardest world: six of the nine models complete none of its tasks, the best completes 33%, and the mean across models is 8.5%. The renovation (18.5%) and the sanatorium (19.1%) follow; in both, completing a task requires keeping track of a state that changes, through the agent's own work or through a pursuer.

Strong agents are economical. GPT-6 Astra spends 61% of its step budget and Gemini 4 Argon 67%; Claude Opus 5.5 spends 75% and the remaining models 80–93% (Table 7). Across models, budget use correlates with success at $r = -0.91$: weak agents do not fail for lack of steps but spend them without converging.

Table 3: **Outcome per model and test seed on the two topology-dependent tasks and their ordinary-floor controls** (one symbol per episode; three seeds for the top three models). ✓ success; ✗ failure; ∅ the agent declared a reachable goal impossible. On `tether_ctrl` the rope is too short, so declaring impossible with the right reason is the success.

Task	Astra	Gemini	Opus	Muse	DeepSeek	Kimi	Qwen	Grok	Doubao
`chalk` (impossible floor)	✓✓✓	✓✓∅	∅∅∅	✗	∅	∅	∅	✗	∅
`chalk_ctrl` (ordinary floor)	✓✓✓	✓✓✓	✓✓✓	✗	✗	✓	✗	✓	✗
`tether` (impossible floor)	✓✓✓	✓✓∅	∅✓✓	∅	∅	∅	∅	✗	∅
`tether_ctrl` (ordinary floor)	✓✓✓	✓✓✓	✓✓✓	✓	✓	✗	✓	✗	✓

Robustness across seeds. Performance is highly consistent across independent test seeds (Table 9): GPT-6 Astra completes 33, 33 and 32 of the 59 tasks across its three seeds, Gemini 4 Argon completes 34, 33 and 29, and Claude Opus 5.5 completes 25, 24 and 25. The two leading models perform within seed-to-seed variation of one another, with Gemini 4 Argon leading on the primary seed and their task-level bootstrap confidence intervals overlapping almost entirely.

> *Two models form a clear top tier, the rest fall off a cliff, and reference-frame shifts in the detective's office pose the steepest barrier.*

5 How Agents Fail in Action

To understand when and why agents fail, we examine their trajectories and the reasons they state for their terminal decisions, in the spirit of analysing models' self-explanations (Yang et al., 2025a). Because each turn records the action, the feedback and, at `declare_impossible`, the agent's reason in its own words, we can ask at which step of the loop a failure arises: in perceiving the change, in updating the model, or in acting on it.

Of the 885 episodes, 568 fail. Assigning each failure to the way it ended, 133 end with the agent declaring the task impossible, 99 with the agent announcing that it is done when it is not, 89 with the agent caught by a pursuer, and 85 on the obstacle course after repeated falls (76 running out of steps, 9 giving up); the remaining 162 run out of steps in the other worlds, break a task rule, or end early or with an audit flag. The first two endings are decisions the agent makes about the state of the world, and we examine them first.

5.1 Seeing the Anomaly, Misreading It

The corridor's floor cannot exist in Euclidean space, and two of its tasks hinge on exactly that. In `chalk`, the agent must carry a box from the B1 landing of the west stairs down to the B3 pump room, but the west stairs lead back to B1 every time; B3 is reachable by another route. In `tether`, a 50 m rope tied at the lift lobby must reach the archive; on an ordinary floor the shortest route is 59 m and the right answer is to declare the task impossible, but on the impossible floor the ring's connectivity brings the archive within reach.

Local competence, global failure. Agents handle the impossible floor well when local, reactive navigation suffices: the door-wedge and pursuit tasks succeed in 10 and 11 of their 15 episodes. The two tasks that require reasoning about the floor's global connectivity divide the models (Table 3). GPT-6 Astra succeeds in all six of its episodes on the impossible floor, Gemini 4 Argon in four, Claude Opus 5.5 in two and the other six models in none. Over all models, these tasks succeed in 12 of 30 episodes on the impossible floor and in 24 of 30 on the ordinary one.

The anomaly is perceived. What separates success from failure is not perception. Figure 6a contrasts two agents on the same seed of `chalk`. Both mark the wall with chalk, walk the stairs and see their own mark come round again. Every one of the eight episodes that give up on `chalk` reports exactly this in its stated reason[1]: the stairs keep returning to the same landing, where the agent's own chalk mark is still there; six of the eight call the stairwell "a closed loop". Their perception and memory of the change are correct.

The update is not acted on. Having recorded the anomaly, failing agents treat it as proof that the goal is unreachable once the landing door fails to open for them, rather than as a reason to leave the stairwell and look for another route, which is what Gemini 4 Argon does. On `tether`, failing agents follow the corridor until the rope pulls taut at about 45 m, some also try the stairs, and they declare the archive out of reach, while the leaders find a route that reaches it (Gemini 4 Argon arrives with 38 m of rope paid out). Two of them state the assumption outright: any detour "would only add rope length", and the archive is out of reach "whichever route" is taken. In total, 15 of the 18 failures on these tasks are confident declarations that a reachable goal is impossible, while no agent declares the ordinary-floor `chalk_ctrl` impossible. We call this the *impossibility shortcut*: when observations contradict a prior about how space works, agents resolve the conflict by declaring the world impossible instead of revising their plan. Humans navigating environments with such "wormholes" abandon Euclidean metric assumptions in favour of a topological graph (Warren et al., 2017); only the leading models make this shift, whereas weaker models cling to Euclidean priors and declare a solvable world impossible.

Beyond the corridor. The impossibility shortcut is not confined to the impossible floor. Across the corridor, sanatorium and office tasks that can be completed, agents declare the task impossible in 104 of 495 episodes (21%), from 7 of 99 for Gemini 4 Argon to 15 of 33 for Doubao Seed 2.1 Pro, and the rate falls with success across models ($r = -0.54$; Table 8).

> *Agents perceive and record change correctly; they fail to revise their plan around it and take the shortcut of declaring the world impossible.*

5.2 Losing Track of One's Own Changes

When the change is the agent's own doing, the world offers no external signal of it: no progress bar marks a cleaned floor, and nothing highlights a wardrobe the agent pushed aside. Keeping track falls entirely on the agent's working memory.

Premature completion. Agents routinely declare work finished that is not (Figure 6b; per-model counts in Table 8). In the renovation, 72 of 150 episodes end with a `stop` while work-order items remain incomplete, and every model does so at least twice; in the office, 27 of 120. The control that reports the remaining stains per room raises cleaning from 1 to 9 successes out of 15 episodes, and starting with the wall already removed raises the knock-through task from 2 to 9 (Figure 8), locating the deficit in tracking completion and updating after one's own changes rather than in cleaning or demolition.

Rules are read and obeyed. The same agents rarely break explicit constraints. Across 150 renovation episodes, a load-bearing wall marked on the client's plan is demolished once; wrong sealed boxes are opened in 10 of 120 office episodes. What fails is not reading the rules but keeping the ledger of a world in progress.

> *Agents follow explicit rules but lose the ledger of their own changes, and stop before the work is done.*

[1] Agents wrote their reasons in Chinese, the language of the task cards; quotations are our translations.

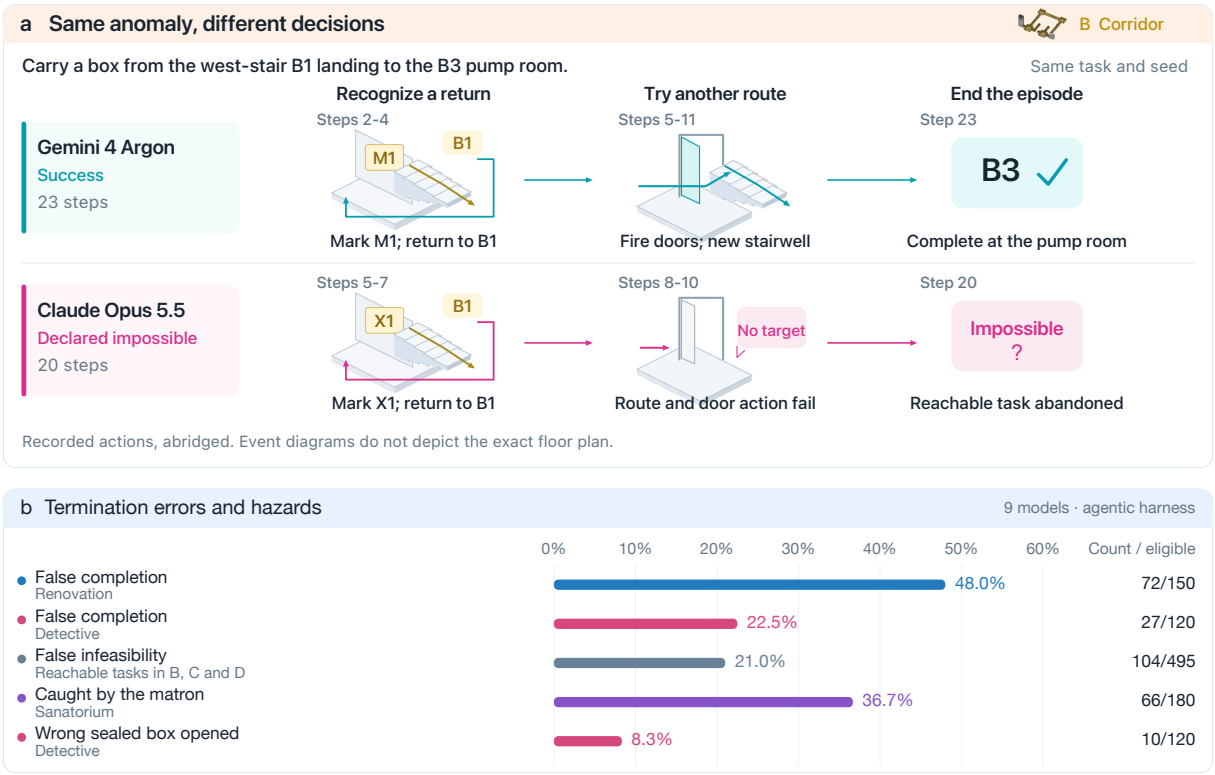

Figure 6: **How agents fail.** (a) Same anomaly, different decisions (`loop/chalk`, one test seed). Both agents use chalk to detect that the stairs loop; perceiving and remembering the change succeed. Gemini 4 Argon leaves the stairwell through its fire door and finds another way down; Claude Opus 5.5, once its attempt at the landing door fails, declares a reachable task impossible. (b) Termination errors and hazards across all nine models and 885 episodes. Each rate is a count over its own set of eligible episodes, so rates must not be added; the detective rates exclude the two high-display-case tasks, whose feasibility depends on the seed. Per-model counts are in Table 8.

5.3 Acting Against Moving Parts

Where the world moves on its own, agents need a forward model of something they do not control. In the sanatorium, agents are caught by the matron in 66 of 180 episodes (13, 13 and 17 of 36 for the top three models; Table 8), and the top three are caught 13, 13 and 21 times in all when the corridor pursuit tasks are included; only the top three ever complete the escort out through the mortuary tunnel, GPT-6 Astra on all three seeds. On the course, success ranges from 8% to 86% across models, and the hammer corridor, whose four hammers swing with different periods, is passed in only 2 of 15 episodes. These two worlds expose the widest gaps between the leaders and the rest.

Forward simulation of things that move on their own separates the leaders from the rest.

6 What Helps Agents Keep Up?

Having located the failures, we ask which kinds of support close them: memory the agent can write to, the removal of single difficulties, and direct assistance with the map and the plan.

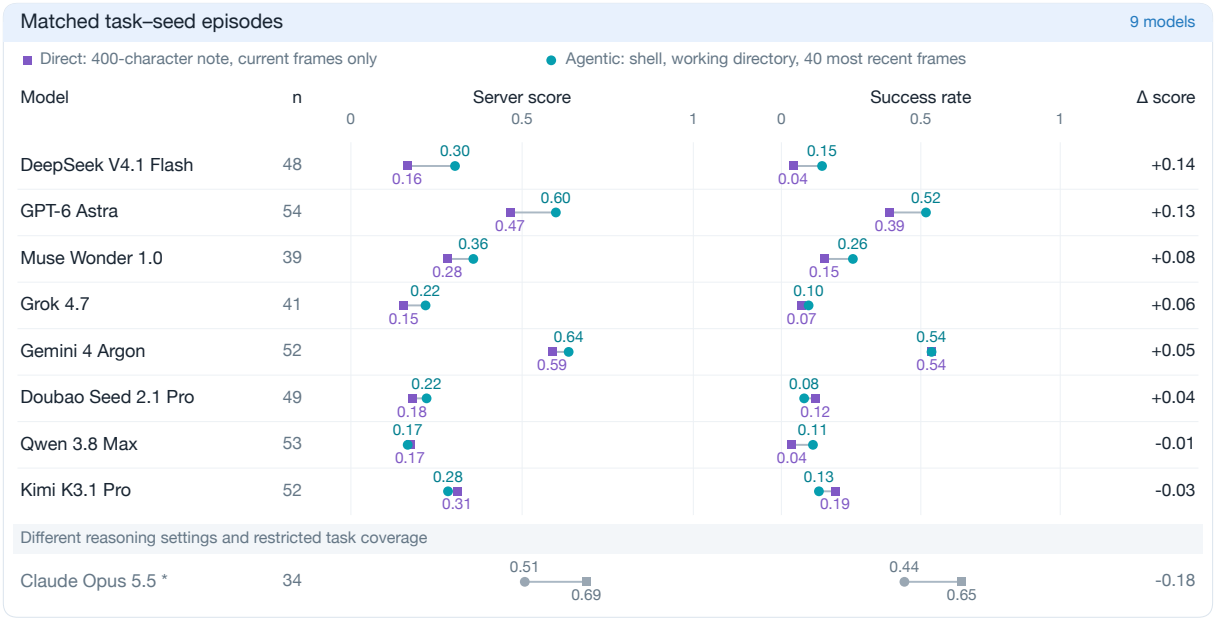

Figure 7: **Direct vs. agentic harness** on matched (task, seed) episodes: server score (left; an audit flag scores zero) and success rate (right); Δ = agentic $-$ direct server score, computed before rounding. The agentic harness provides a shell, a writable working directory, and a 40-frame visual context window, whereas the direct harness provides the current frames and a 400-character rolling note. *Claude Opus 5.5 was evaluated on 34 tasks at `max` reasoning effort in the direct harness versus `xhigh` in the agentic harness, and is listed separately.

6.1 External Workspace vs. Rolling Memory

The direct harness restricts the agent to a 400-character rolling note and the current frames; the agentic harness equips it with a shell, a persistent working directory for structured notes, and a 40-frame visual history. On matched episodes (Figure 7), the agentic harness improves the server score of six of nine models, led by DeepSeek V4.1 Flash ($0.16 \rightarrow 0.30$) and GPT-6 Astra ($0.47 \rightarrow 0.60$, lifting task success from 39% to 52%). Externalising memory into a writable workspace thus provides a clear advantage. Crucially, however, all failure modes analysed in §5 were measured with this agentic workspace active—demonstrating that premature termination and the impossibility shortcut are fundamental bottlenecks in spatial state revision rather than mere artifacts of a small text scratchpad.

> *A writable workspace improves most models, yet cannot by itself calibrate termination or force plan revision around an anomaly.*

6.2 Removing One Difficulty at a Time

The 15 counterfactual controls isolate the exact cost of each spatial demand (Figure 8, Table 10). Removing demands on *world-state revision* and *completion tracking* yields dramatic gains: replacing the non-Euclidean floor with an ordinary layout raises `chalk` from 5 to 11 successes out of 15 ($+40$ pp) and `tether` from 7 to 13 ($+40$ pp); providing per-room stain counts raises `sweep` from 1 to 9 ($+53$ pp); and starting with the partition already demolished raises `open_plan` from 2 to 9 ($+47$ pp). Pooled across all seven corridor pairs, restoring an ordinary floor lifts success from 54 to 65 of 105 episodes, with the gain concentrated in tasks that hinge on global connectivity. By contrast, controls that only simplify *local perception or static naming*—such as replacing relational room descriptions with door plates (`paint_rel`, `repaint_ref`)—

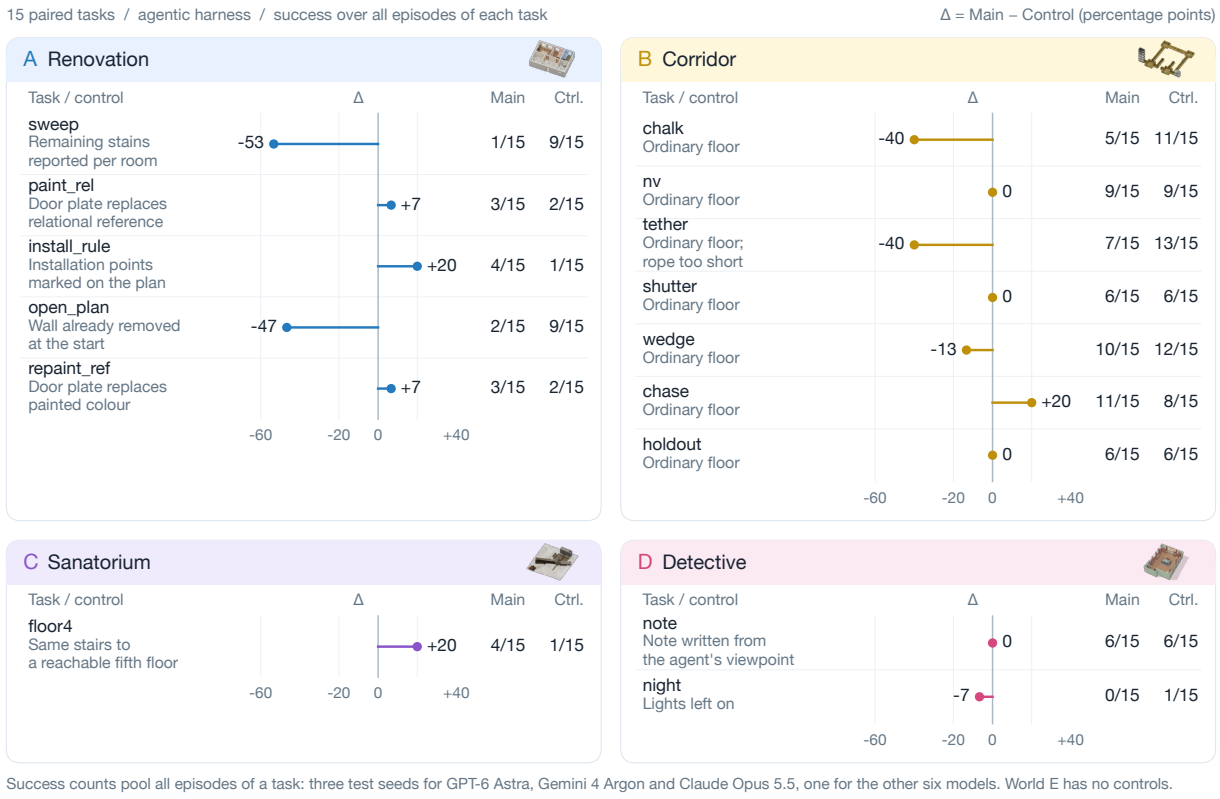

Figure 8: **Counterfactual controls** (agentic harness, nine models). For each of the 15 pairs: the difficulty the control removes, Δ = main − control in success rate (percentage points; left of zero: the control succeeds more often), and the successes out of all episodes of each task (three test seeds for the top three models, one for the other six). World E has no controls. Table 10 adds progress.

shift success by a single episode, ruling out local visual recognition as the primary bottleneck. Meanwhile, on the sanatorium's `floor4`/`floor5` pair, agents succeed more often when the fourth floor is genuinely blocked ($4/15$) than when the same stairs reach a feasible fifth floor ($1/15$), directly reflecting their bias toward premature `declare_impossible` calls.

6.3 Stratified Assistance

We rerun the 44 main tasks on their first test seed with more of the problem solved for the agent, for all nine models through the same harness (396 episodes per condition, paired with the first-seed episodes of §4). With `+map`, the agent receives at the start a ground-truth plan of the world as it is at that moment: the layout, its own position and heading, where the task's objects lie, and where stairs and mechanisms lead; the plan is not updated as the world changes. With `+plan`, it also receives a correct plan in words: the route through named places, the order of the work items, which box and bookcase to open, or how to time each obstacle on the course; the routes in the sanatorium leave the matron to the agent. Writing s_0, s_1 and s_2 for the success rates without help, with the map and with the plan, we split the baseline failures $1 - s_0$ into the share the map removes, $(s_1 - s_0)/(1 - s_0)$, which we attribute to building the spatial model; the share the plan removes on top of it, $(s_2 - s_1)/(1 - s_0)$, attributed to planning; and the share left with both, $(1 - s_2)/(1 - s_0)$, attributed to execution (Table 4).

Table 4: **Stratified assistance** (agentic harness; the 44 main tasks on their first test seed, nine models). Success rate without help, with the true map (`+map`) and with the map and a correct plan (`+plan`), and the baseline failures split into the share the map removes (spatial model), the share the plan removes on top of it (planning) and the share left with both (execution); the three shares sum to 100%. Per-model results are in Table 11.

		Success (%)			Baseline failures (%)		
World	n	No help	`+map`	`+plan`	Spatial model	Planning	Execution
A Renovation	45	15.6	28.9	48.9	15.8	23.7	60.5
B Corridor	72	33.3	83.3	87.5	75.0	6.2	18.7
C Sanatorium	99	21.2	26.3	45.5	6.4	24.4	69.2
D Detective	72	6.9	30.6	41.7	25.4	11.9	62.7
E Course	108	37.0	59.3	54.6	35.3	−7.4	72.1
All	396	24.5	46.7	55.3	29.4	11.4	59.2

The map nearly doubles success, while dynamic execution retains most failures. Overall success rises from 24.5% without assistance to 46.7% with `+map` and 55.3% with `+plan` (reaching 86.4% for GPT-6 Astra), while 59.2% of baseline failures persist even with both forms of assistance. The decomposition across worlds is diagnostic (Table 4). In the corridor, `+map` alone eliminates 75.0% of baseline failures (33.3% → 83.3%): once given the true non-Euclidean connectivity on the map, all nine models solve both `chalk` and `tether`, compared with only two of nine unassisted—confirming that the bottleneck in §5.1 lies in revising the spatial model from anomalous observations rather than in following a route. In the renovation flat, the ordered work-item sequence of `+plan` resolves a larger share of failures than the static map (23.7% vs. 15.8%, lifting success to 48.9%), directly addressing the lost completion ledger of §5.2. By contrast, in the sanatorium and on the obstacle course, 69.2% and 72.1% of baseline failures remain even under `+plan`. In these worlds, success depends on closed-loop timing against a moving pursuer or periodic obstacles: on the course, where the linear track has no route ambiguity and timing must be grounded in the live 4-frame burst, verbal timing advice (`+plan`, 54.6%) offers no advantage over the map alone (`+map`, 59.3%). Across models, the share of failures attributable to execution scales inversely with capability, dropping from 28.6% for GPT-6 Astra to 55.0–73.0% for the six weaker models (Table 11).

Ground-truth maps resolve non-Euclidean routing, and verbal plans resolve multi-step ordering, but closed-loop execution against moving threats and obstacles remains the decisive bottleneck.

7 Related Work

Spatial intelligence in MLLMs. Building on the reasoning abilities of large language models, MLLMs are increasingly evaluated on spatial understanding: from single images (Chen et al., 2024; Cheng et al., 2024; Ma et al., 2025), multiple views (Yeh et al., 2026; Wang et al., 2026a) and egocentric video (Yang et al., 2025a; Lin et al., 2025; Zhu et al., 2026). These works established that MLLMs build strong local but weak global spatial models and that explicit cognitive maps help (Yang et al., 2025a; Wang et al., 2026a). Their observations are collected in advance, and the world does not respond to the model. Theory of Space (Zhang et al., 2026a) and SpatialScout (Zhang et al., 2026b) let agents gather their own observations in static scenes. Space Track extends this line to worlds that change while the agent acts and measures understanding through what the agent achieves.

Table 5: **Comparison with related benchmarks.** ✓ supported; △ partially; ✗ not supported.

Benchmark	Closed-loop action	First-person 3D	World changes mid-episode	Agent changes layout	Adversary	Non-Euclidean topology	Counterfactual controls
VSI-Bench (Yang et al., 2025a)	✗	✓	✗	✗	✗	✗	✗
MindCube (Wang et al., 2026a)	✗	△	✗	✗	✗	✗	✗
OST-Bench (Lin et al., 2025)	✗	✓	✗	✗	✗	✗	✗
Theory of Space (Zhang et al., 2026a)	✓	✓	△	✗	✗	✗	△
SpatialScout (Zhang et al., 2026b)	✓	✓	✗	✗	✗	✗	✗
SpatialAct (Liu et al., 2026)	✓	✗	✓	△	✗	✗	✗
EmbodiedBench (Yang et al., 2025b)	✓	✓	△	✗	✗	✗	✗
ChangeVLN (Kang and Kim, 2026)	✓	✓	✓	✗	✗	✗	✗
VideoGameBench (Zhang et al., 2025)	✓	△	✓	✗	✓	✗	✗
BALROG (Paglieri et al., 2025)	✓	✗	✓	△	✓	✗	✗
Space Track (ours)	✓	✓	✓	✓	✓	✓	✓

Embodied and interactive agents. Instruction-following and household benchmarks evaluate long-horizon action in simulated homes (Shridhar et al., 2020; Padmakumar et al., 2022; Srivastava et al., 2022; Li et al., 2024; Yang et al., 2025b); agents move objects, but the structure of the home is fixed and progress is observable. SpatialAct (Liu et al., 2026) evaluates multi-turn object edits judged from top-down and isometric renders, and iVISPAR (Mayer et al., 2025) sliding-tile planning across 2D, 3D and text. Space Track acts from first-person views only, lets the agent change the connectivity of the space, and provides no progress signal.

Navigation under change. Vision-and-language navigation (Anderson et al., 2018) has been extended to blocked routes and outdated instructions (Liu et al., 2025; Kang and Kim, 2026; Yu et al., 2026), wayfinding by signs and floor plans (DeFazio et al., 2025; Zimmerman et al., 2026a,b) and real-city exploration (Wang et al., 2026b, 2025). There the change is imposed by the environment and the task is to reach a goal. Space Track adds mechanisms the agent operates, topologies that violate Euclidean geometry (Warren et al., 2017), adversaries, and infeasibility judgements audited against what the agent actually saw.

Games and physical reasoning. VideoGameBench (Zhang et al., 2025) and BALROG (Paglieri et al., 2025) evaluate agents on games, and VideoGameBench introduced pausing the game during inference, which we adopt. Commercial games mix many skills and admit no controlled variants; physical reasoning benchmarks (Bakhtin et al., 2019; Bear et al., 2021) pose questions to an observer. Space Track builds each world so that a single difficulty can be removed, and asks the agent to act on its predictions.

8 Discussion and Future Work

We study how multimodal agents perceive, update and act in 3D worlds that change while they act, by building Space Track and diagnosing nine frontier models across 885 main evaluation episodes and 792 stratified-assistance episodes. A steep capability cliff separates GPT-6 Astra (55.4%) and Gemini 4 Argon (54.2%) from the remaining models, and shifting reference frames in the detective's office poses the steepest barrier. Trajectory-level analysis and counterfactual controls locate the primary bottleneck after initial perception: agents detect a looping stairwell and record it in their own words, yet default to declaring a reachable goal impossible; they obey explicit constraints almost without fail, yet lose the ledger of their own renovations and terminate prematurely. Externalising memory into a writable workspace improves six of nine models, and providing a ground-truth map eliminates three quarters of non-Euclidean corridor failures, while closed-loop execution against patrolling adversaries and periodic obstacles remains challenging even when a correct

verbal plan is supplied. Together, these findings point to three priorities for embodied foundation models: training in closed-loop environments where revising a plan around a spatial anomaly is rewarded over premature exit; structured working-memory mechanisms that maintain an explicit ledger of self-induced state changes; and calibrated termination policies that treat `stop` and `declare_impossible` as verifiable world-state judgements. Space Track reframes spatial evaluation from "can the model recall a static space?" to "can the model keep its picture of space current while acting within it?".

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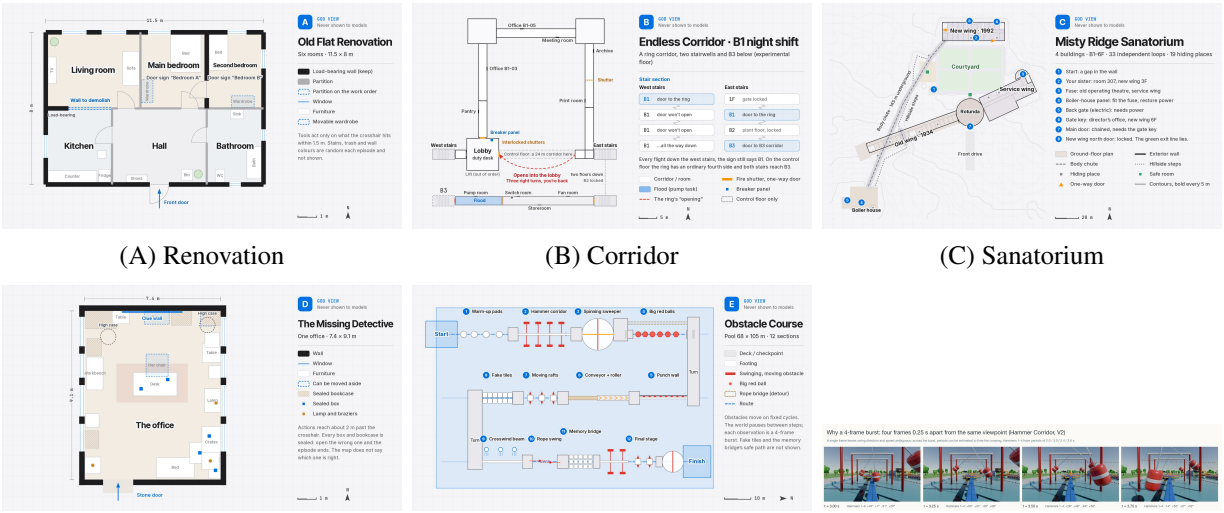
A Problem Formulation

Following the standard formulation of vision-driven agents as partially observable decision processes (Yang et al., 2025b), an episode of Space Track is a tuple $(\mathcal{S}, \mathcal{A}, \Omega, \mathcal{T}, \mathcal{O}, L, \mathcal{R})$. The state $s_t \in \mathcal{S}$ is never observed by the agent; we decompose it as $s_t = (G_t, X_t, E_t)$, where G_t is the navigable connectivity graph, X_t the state of objects and of the task (stains, paint, fixtures, moved furniture, items held, mechanisms), and E_t the state of entities the agent does not control (a pursuer, moving obstacles). Observations $o_t \in \Omega$ are first-person frames with a one-sentence feedback, $o_t \sim \mathcal{O}(\cdot \mid s_t)$. L is the task card, and $\mathcal{R}$ the task's scoring function.

The transition $\mathcal{T}(s_{t+1} \mid s_t, a_t)$ changes the world through the five channels of §2: self-induced changes of G and X (A), agent-triggered mechanisms changing G (B), exogenous dynamics of E (C), changes of X the agent must later undo under instructions given in another reference frame (D), and periodic dynamics of E (E). The agent keeps a history $h_t = (o_0, a_0, \ldots, o_t)$ and selects actions by a policy $\pi(a_t \mid L, h_t)$. Two terminal actions are judged: `stop`, which is correct if and only if s_t satisfies L, and `declare_impossible`, which is correct if and only if no policy can reach a state satisfying L from s_t. An episode also ends when the step budget is exhausted or, in worlds that offer it, on `give_up`. The objective is to maximise $\mathbb{E}[\mathcal{R}(s_\tau, L)]$ at the terminal step τ. A counterfactual control keeps L and $\mathcal{R}$ and removes one channel of change or one demand on the agent, so the task–control gap measures the cost of that demand.

B World Details



(A) Renovation (B) Corridor (C) Sanatorium

(D) Detective (E) Course Course: a 4-frame burst

Figure 9: **Top-down views of the five worlds**, for readers only: agents never see a map in the standard condition. Bottom right: one observation on the course is a burst of four frames 0.25 s apart, from which the hammers' periods must be inferred.

Renovation. Body: a 25 cm circle colliding with walls and furniture; walking 1.25 m/s, turning 120°/s. Actions: `walk`, `back`, `sidestep`, `turn`, `look`, `wait`, `plan`, and the tools `clean`, `pick_up`, `dispose`, `paint(colour)`, `demolish`, `install(fixture)`, `move_aside`, `look_closer`, each acting within 1.5 m under the crosshair, plus `stop`, `declare_impossible` and `give_up`. At

reset the agent also receives the client's floor plan with load-bearing walls in black. An optional photoreal mode replaces the browser frames with Blender Cycles renders of the same state (about 3–4 s per frame on CPU) with identical rules and scoring.

Corridor. Junction-level navigation plus `use(name)` for switches at the agent's stop, `wait(s)`, and task tools: `mark(text)` (chalk), `night_vision` (one battery bar per action while on), `rewind(n)` (the frame from n steps earlier). The pursuer walks the floor graph toward the agent slightly faster than it, is stopped by lowered shutters and slowed by fire doors.

Sanatorium. Junction-level navigation (`forward`, `turn_left`/`right` by $45°$, `turn_around`) plus `interact`, `run`, `wait(s)`, `flashlight`, `take_elevator`, `stop` and `declare_impossible(reason)`. The matron's rules (hearing by walking distance, sight at 30 m with the flashlight on, locker searches, safe rooms she cannot enter) are stated in the task card. Optimal solutions take 8–38 steps.

Detective. Parametric movement as in the renovation, plus `inspect`, `move_aside`, `toggle_light`, `pick_place` and `mark_anomaly`, acting within about 2.25 m under the crosshair (2.1 m horizontally for high display cases). Opening the wrong sealed box or bookcase ends the episode; marking an object as moved that was not counts as a false alarm.

Course. Actions `wait`, `walk`, `run`, `back`, `sidestep`, `turn`, `jump(m)` (up to 3.2 m, inheriting the platform's velocity), `duck`, `grab`, `release`, `look`, `plan` and `give_up`. Falling into the pool returns the agent to the last checkpoint after 3 s with time still running, so recovery from failure is observable. Each section has its own step budget and a 300 s limit.

C Scoring

Success is decided by each task's own checker; an audit flag turns a success into a failure. *Progress* is 1 for a success and otherwise graded as follows. Renovation: the mean completion of work-order items minus violation penalties (out-of-scope paint, wrong demolition, extra fixtures, furniture not pushed back); zero if a load-bearing wall is demolished. Office: the mean progress of task items; the episode ends on a broken seal. Course: each task is one section, so progress is 1 if it is passed and 0 otherwise. Corridor and sanatorium: $0.9\times$ the covered share of the shortest-path distance on the floor graph from the start (or the last goal reached) to the next goal, plus goals already reached, divided by the number of goals; survival tasks use the share of the required time survived; tasks that cannot be done use the server's score for the stated reason. Declaring impossible a task that can be done halves progress, and an audit flag sets it to zero. A model's progress is the mean over tasks of the task's mean over seeds.

D Harness Details

Agentic harness task prompt (translated from the Chinese original).

> You are taking part in the Space Track evaluation: complete a task in a 3D world from a first-person view. No one will answer your questions and you do not need to ask for permission; start right away. You can only interact with the world through the Space Track tools. First call `get_task` to see the task card (task, rules, available actions, step budget), the task attachment and the opening frame. Then call `act` repeatedly; it returns a one-sentence feedback and the first-person frame after the action. `plan`

can take several actions at once, each counting as a step. The world is paused while you think; only actions advance game time. Keep acting until `act` tells you the episode is over: when the task is complete, end as the task card requires (usually `stop`); when you are sure it cannot be done, use `declare_impossible` and state the reason. Do not stop midway. The current directory `/work` is empty; you may take notes and write files here. There is no network and no other material about this episode.

Direct harness system prompt (translated).

You control a first-person character in the Space Track evaluation and complete a task in a 3D world. You learn about the world only through frames and one sentence of feedback per step; the world is paused while you think. Each turn you see the task card, your notes, the record of every step so far (action and feedback, and your judgement for the most recent steps), and the current first-person frames. Reply each turn with exactly one JSON object: `{"notes": ..., "reason": ..., "action": ..., "arg": ...}`. *notes* is your memory of the world, given back to you next turn and replaced by the version you write now: layout, where you have been, what is done and what is not, changes you observed; at most 400 characters. *reason* is your judgement of the current frame and situation and why you choose this action; at most 150 characters. *action* must be a name from the action list. *arg* is a number, a string or a list as the action requires, or null. When the task is complete, end as the task card requires (usually `stop`); when you are sure the task cannot be done, use `declare_impossible` with the reason in *arg*.

Answer handling. Answers stopped by a provider-side filter or returned empty are requested again without counting against the model. In the direct harness, an answer split over several JSON objects is merged, and broken JSON is read by key; after two invalid answers in one step the episode ends and the partial score stands. In the agentic harness, five consecutive unknown actions end the episode. Provider outages are infrastructure errors: the episode is discarded and rerun.

E Models

Table 6: **Evaluated models and reasoning settings.**

Model	Reasoning setting
Gemini 4 Argon	thinking level `high`
GPT-6 Astra	reasoning effort `xhigh`
Claude Opus 5.5	reasoning effort `xhigh`
Muse Wonder 1.0	reasoning effort `max`
DeepSeek V4.1 Flash (1017)	reasoning effort `max`
Kimi K3.1 Pro	thinking enabled
Grok 4.7	reasoning effort `xhigh`
Doubao Seed 2.1 Pro	reasoning effort `max`
Qwen 3.8 Max (0902)	reasoning effort `max`

F Additional Results

Table 7: **Full results** (agentic harness, 885 episodes; Figure 5). Success rate, successes over episodes, progress, success rate per world (%), and the share of the step budget used (lower is more efficient). Best in bold.

#	Model	Overall Succ. %	Overall k/n	Overall Prog.	A Renov.	B Corr.	C Sanat.	D Detect.	E Course	Budget used ↓
1	GPT-6 Astra	**55.4**	98/177	**70.2**	20	**96**	**28**	27	**86**	**0.61**
2	Gemini 4 Argon	54.2	96/177	69.8	**43**	91	22	**33**	67	0.67
3	Claude Opus 5.5	41.8	74/177	58.8	33	62	22	17	64	0.75
4	Muse Wonder 1.0	25.4	15/59	45.6	40	27	17	0	42	0.83
5	DeepSeek V4.1 Flash	15.3	9/59	35.2	0	27	25	0	17	0.85
6	Kimi K3.1 Pro	13.6	8/59	39.3	10	13	25	0	17	0.81
7	Qwen 3.8 Max	11.9	7/59	24.9	10	7	17	0	25	0.80
8	Doubao Seed 2.1 Pro	8.5	5/59	32.5	0	13	8	0	17	0.85
9	Grok 4.7	8.5	5/59	28.9	10	13	8	0	8	0.93

Table 8: **Termination and hazards** (agentic harness, 885 episodes; counts over eligible episodes). False completion: `stop` before the work is complete. False infeasibility: `declare_impossible` on a task that can be done (corridor, sanatorium and office). Caught: ended by the matron. Seal: wrong sealed box opened.*

Model	False completion		False	Caught	Seal
	Renovation	Detective	infeasibility	(Sanat.)	broken
GPT-6 Astra	17/30	0/24	20/99	13/36	3/24
Gemini 4 Argon	15/30	9/24	7/99	13/36	0/24
Claude Opus 5.5	15/30	9/24	21/99	17/36	0/24
Muse Wonder 1.0	2/10	4/8	4/33	6/12	1/8
DeepSeek V4.1 Flash	6/10	0/8	10/33	2/12	2/8
Kimi K3.1 Pro	5/10	1/8	13/33	5/12	1/8
Qwen 3.8 Max	6/10	3/8	11/33	2/12	2/8
Doubao Seed 2.1 Pro	4/10	1/8	15/33	4/12	1/8
Grok 4.7	2/10	0/8	3/33	4/12	0/8
All models	72/150	27/120	104/495	66/180	10/120

*The two office tasks about a high display case are excluded from the office columns, because whether the case can be opened depends on the seed.

Table 9: **Seed robustness** (agentic harness). Tasks completed on each of the three test seeds, and the overall success rate with a bootstrap 95% interval that resamples tasks (each with its three seeds).

Model	Seed 1	Seed 2	Seed 3	Success % [95% CI]
GPT-6 Astra	33/59	33/59	32/59	55.4 [45.2, 66.1]
Gemini 4 Argon	34/59	33/59	29/59	54.2 [44.1, 64.4]
Claude Opus 5.5	25/59	24/59	25/59	41.8 [31.6, 51.4]

Table 10: **All 15 counterfactual controls** (agentic harness). Successes over all 15 episodes of the main task and of its control (three test seeds for the top three models, one for the other six), Δ = main $-$ control in success rate (percentage points), and the mean progress of each across the 15 episodes.

W	Task	Difficulty in the main task	Control	Main	Ctrl	Δ	Prog. main	Prog. ctrl
A	`sweep`	progress feedback	remaining stains reported per room	1/15	9/15	-53	0.70	0.80
A	`paint_rel`	relational reference	door plate instead of "room next to the bathroom"	3/15	2/15	+7	0.85	0.82
A	`install_rule`	rule grounding	install points marked on the plan	4/15	1/15	+20	0.61	0.60
A	`open_plan`	self-induced layout change	wall already removed at start	2/15	9/15	-47	0.67	0.60
A	`repaint_ref`	self-changed landmark	door plate instead of the colour the agent painted	3/15	2/15	+7	0.57	0.55
B	`chalk`	repeating stairwell	ordinary floor	5/15	11/15	-40	0.37	0.89
B	`nv`	impossible ring, darkness	ordinary floor	9/15	9/15	+0	0.72	0.65
B	`tether`	impossible ring, metric reach	ordinary floor (rope too short)	7/15	13/15	-40	0.50	0.87
B	`shutter`	interlocked shutters	ordinary floor	6/15	6/15	+0	0.41	0.48
B	`wedge`	one-way door	ordinary floor	10/15	12/15	-13	0.77	0.83
B	`chase`	pursuit on impossible ring	ordinary floor	11/15	8/15	+20	0.90	0.81
B	`holdout`	survival on impossible ring	ordinary floor	6/15	6/15	+0	0.61	0.61
C	`floor4`	infeasibility (burnt 4F)	same stairs to a reachable 5F	4/15	1/15	+20	0.39	0.29
D	`note`	perspective taking	note written from the agent's side	6/15	6/15	+0	0.60	0.47
D	`night`	memory in the dark	lights left on	0/15	1/15	-7	0.50	0.58

Table 11: **Stratified assistance per model** (agentic harness; the 44 main tasks on their first test seed; Table 4). Success rate without help, with the true map and with the map and a correct plan, and the baseline failures split into the shares removed by the map, removed by the plan on top of it, and left with both.

		Success (%)			Baseline failures (%)		
Model	n	No help	`+map`	`+plan`	Spatial model	Planning	Execution
GPT-6 Astra	44	52.3	70.5	86.4	38.1	33.3	28.6
Gemini 4 Argon	44	52.3	63.6	77.3	23.8	28.6	47.6
Claude Opus 5.5	44	38.6	72.7	77.3	55.6	7.4	37.0
Muse Wonder 1.0	44	25.0	50.0	54.5	33.3	6.1	60.6
DeepSeek V4.1 Flash	44	15.9	22.7	38.6	8.1	18.9	73.0
Kimi K3.1 Pro	44	9.1	50.0	50.0	45.0	0.0	55.0
Qwen 3.8 Max	44	13.6	36.4	40.9	26.3	5.3	68.4
Doubao Seed 2.1 Pro	44	6.8	22.7	38.6	17.1	17.1	65.9
Grok 4.7	44	6.8	31.8	34.1	26.8	2.4	70.7
All	396	24.5	46.7	55.3	29.4	11.4	59.2

Table 12: **Progress per task and model** (agentic harness; mean over test seeds, three for the top three models), and successes over all episodes of the task.

W	Task	Astra	Gemini	Opus	Muse	DeepSeek	Kimi	Qwen	Grok	Doubao	Succ.
A	sweep	0.55	0.98	0.96	0.82	0.44	0.44	0.17	0.41	0.69	1/15
A	sweep_hud	0.83	1.00	1.00	1.00	0.49	0.97	0.00	0.52	0.52	9/15
A	paint_rel	0.92	0.79	0.92	1.00	0.75	0.75	0.75	0.75	0.88	3/15
A	paint_plate	0.88	0.75	0.96	0.75	0.75	0.75	0.75	0.75	0.75	2/15
A	install_rule	0.62	0.93	0.74	0.00	0.62	0.62	1.00	0.00	0.00	4/15
A	install_marked	0.62	0.86	0.49	0.40	0.62	0.62	0.62	0.62	0.28	1/15
A	open_plan	0.68	0.86	0.78	0.56	0.52	0.44	0.63	0.41	0.48	2/15
A	open_plan_pre	0.67	1.00	0.33	1.00	0.00	1.00	0.00	1.00	0.00	9/15
A	repaint_ref	0.56	0.58	0.64	1.00	0.33	0.50	0.50	0.58	0.33	3/15
A	repaint_plate	0.36	0.63	0.72	0.42	0.50	0.83	0.33	0.50	0.58	2/15
B	chalk	1.00	0.67	0.00	0.58	0.00	0.00	0.00	0.00	0.00	5/15
B	chalk_ctrl	1.00	1.00	1.00	0.47	0.64	1.00	0.64	1.00	0.64	11/15
B	nv	1.00	1.00	0.70	0.10	1.00	0.66	0.00	0.90	0.05	9/15
B	nv_ctrl	1.00	1.00	0.73	0.20	0.00	1.00	0.20	0.00	0.10	9/15
B	tether	1.00	0.82	0.67	0.00	0.00	0.00	0.00	0.00	0.00	7/15
B	tether_ctrl	1.00	1.00	1.00	1.00	1.00	0.00	1.00	0.00	1.00	13/15
B	pump	1.00	0.67	1.00	0.59	0.00	0.29	0.00	0.00	0.00	8/15
B	shutter	0.67	1.00	0.36	0.00	0.00	0.00	0.07	0.00	0.00	6/15
B	shutter_ctrl	1.00	1.00	0.24	0.00	0.12	0.07	0.07	0.21	0.03	6/15
B	wedge	0.69	1.00	1.00	1.00	1.00	0.18	0.39	0.07	0.78	10/15
B	wedge_ctrl	1.00	1.00	1.00	1.00	1.00	0.29	0.11	0.00	1.00	12/15
B	chase	1.00	1.00	1.00	1.00	0.74	0.90	0.56	1.00	0.36	11/15
B	chase_ctrl	1.00	1.00	0.97	0.90	0.63	0.90	0.36	0.07	0.40	8/15
B	holdout	1.00	1.00	0.31	0.51	0.09	0.48	0.09	0.18	0.84	6/15
B	holdout_ctrl	1.00	0.80	0.52	0.09	0.66	0.63	0.09	0.09	0.57	6/15
C	sister	0.18	0.37	0.00	0.16	0.08	0.01	0.03	0.08	0.52	0/15
C	fuse	0.33	0.10	0.67	1.00	0.00	0.32	0.00	0.49	0.00	4/15
C	phone	0.15	0.10	0.10	0.00	0.00	0.00	0.00	0.00	0.00	0/15
C	records	0.51	0.42	0.23	0.59	0.53	0.59	0.54	0.00	0.38	0/15
C	hill	0.62	0.74	0.50	0.57	0.57	0.77	0.28	0.30	0.77	3/15
C	floor4	0.48	0.48	0.33	0.45	0.00	0.00	1.00	0.45	0.00	4/15
C	floor5	0.00	0.46	0.46	0.00	0.12	1.00	0.24	0.00	0.26	1/15
C	exit_sign	0.74	0.19	0.70	1.00	1.00	1.00	0.00	0.31	1.00	5/15
C	evade	0.44	0.90	0.66	0.07	1.00	0.07	1.00	1.00	0.61	7/15
C	photo	0.35	0.51	0.12	0.15	1.00	1.00	0.00	0.31	0.00	3/15
C	homing	0.72	0.71	0.35	0.00	0.00	0.09	0.00	0.67	0.12	5/15
C	escape	1.00	0.80	0.44	0.39	0.00	0.42	0.00	0.45	0.16	6/15
D	note	0.67	0.83	0.83	0.50	0.00	0.50	0.00	0.50	0.50	6/15
D	note_ctrl	0.33	1.00	0.83	0.00	0.00	0.00	0.00	0.50	0.00	6/15
D	night	0.50	0.61	0.39	0.67	0.67	0.67	0.00	0.50	0.50	0/15
D	night_ctrl	0.72	0.50	0.67	0.67	0.67	0.34	0.00	0.50	0.83	1/15
D	anomaly	0.33	0.21	0.50	0.50	0.50	0.50	0.00	0.50	0.50	0/15
D	infiltrate	0.67	0.92	0.75	0.50	0.25	0.25	0.25	0.25	0.25	4/15
D	restore_relics	0.78	1.00	0.44	0.33	0.33	0.33	0.00	0.00	0.33	5/15
D	cipher	0.17	0.06	0.00	0.00	0.17	0.00	0.00	0.17	0.17	0/15
D	highcase_ne	0.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1/15
D	highcase_w	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0/15
E	s1	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	9/15
E	s2	0.33	0.00	0.33	0.00	0.00	0.00	0.00	0.00	0.00	2/15
E	s3	1.00	0.67	1.00	1.00	1.00	0.00	1.00	0.00	1.00	12/15
E	s4	1.00	0.33	1.00	0.00	0.00	0.00	1.00	0.00	0.00	8/15
E	s5	0.67	1.00	0.33	0.00	0.00	0.00	0.00	0.00	0.00	6/15
E	s6	0.33	0.67	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3/15
E	s7	1.00	0.67	1.00	0.00	0.00	0.00	0.00	0.00	0.00	8/15
E	s8	1.00	0.33	1.00	1.00	0.00	1.00	0.00	0.00	0.00	9/15
E	s9	1.00	1.00	0.67	1.00	0.00	0.00	0.00	0.00	0.00	9/15
E	s10	1.00	1.00	0.33	1.00	0.00	0.00	1.00	1.00	0.00	10/15
E	s11	1.00	1.00	1.00	1.00	1.00	1.00	0.00	0.00	1.00	13/15
E	s12	1.00	0.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4/15